

Language Definition

1 Abstract Syntax

1.1 Values

$$v ::= \begin{array}{l} n \\ | \quad b \\ | \quad C(v_1, \dots, v_n) \end{array}$$

where $n \in \mathbb{Z}$, $b \in \{\mathbf{true}, \mathbf{false}\}$, and C is a constructor name.

1.2 Types

$$T ::= \begin{array}{l} \mathbf{int} \\ | \quad \mathbf{bool} \\ | \quad V \end{array}$$

Variant types are labeled sums:

$$V ::= \langle C_1 : \overline{T}_1, \dots, C_n : \overline{T}_n \rangle$$

where each $\overline{T}_i$ denotes a (possibly empty) sequence of types and C_i are constructor names.

1.3 Expressions

$$e ::= \begin{array}{l} v \\ | \quad e_1 \mathbf{op} e_2 \\ | \quad ! e \\ | \quad x \end{array}$$

where $\mathbf{op} \in \{+, -, *, /, \wedge, \vee, ==, <, >\}$ and x is a variable name.

1.4 Statements

$$s ::= \begin{array}{l} \mathbf{var} \ x : T \\ | \quad x := e \\ | \quad \mathbf{if} \ e \ \{s^*\} \\ | \quad \mathbf{if} \ e \ \{s^*\} \ \mathbf{else} \ \{s^*\} \\ | \quad \mathbf{print} \ e \\ | \quad \mathbf{variant} \ V = C_1(\overline{T}_1) \mid \dots \mid C_n(\overline{T}_n) \end{array}$$

2 Typing Rules

$$\begin{array}{l} \Gamma \vdash e : T \\ \Delta \vdash C(\overline{T}) : V \end{array}$$

2.1 Base Types

$$\frac{}{\Gamma \vdash n : \text{int}} \text{T-INT} \quad \frac{}{\Gamma \vdash \text{true} : \text{bool}} \text{T-BOOL-TRUE} \quad \frac{}{\Gamma \vdash \text{false} : \text{bool}} \text{T-BOOL-FALSE}$$

2.2 Variables

$$\frac{x : T \in \Gamma}{\Gamma \vdash x : T} \text{T-VAR-USE}$$

2.3 Binary Operators

$$\frac{\Gamma \vdash e_1 : \text{int} \quad \Gamma \vdash e_2 : \text{int}}{\Gamma \vdash e_1 + e_2 : \text{int}} \text{T-ARITHMETIC} \quad \frac{\Gamma \vdash e_1 : \text{bool} \quad \Gamma \vdash e_2 : \text{bool}}{\Gamma \vdash e_1 \wedge e_2 : \text{bool}} \text{T-LOGICAL}$$

$$\frac{\Gamma \vdash e_1 : \text{int} \quad \Gamma \vdash e_2 : \text{int}}{\Gamma \vdash e_1 < e_2 : \text{bool}} \text{T-COMPARISON} \quad \frac{\Gamma \vdash e_1 : T \quad \Gamma \vdash e_2 : T}{\Gamma \vdash e_1 == e_2 : \text{bool}} \text{T-EQUALITY}$$

2.4 Unary Operators

$$\frac{\Gamma \vdash e : \text{bool}}{\Gamma \vdash \text{not } e : \text{bool}} \text{T-NOT}$$

2.5 Variants

$$\frac{\Gamma \vdash e_1 : T_1, \dots, e_n : T_n \quad C : (T_1, \dots, T_n) : V \in \Delta}{\Gamma, \Delta \vdash C(e_1, \dots, e_n) : V} \text{T-VARIANT-CONSTRUCTOR}$$

The constructor C must be declared as part of variant V with argument types $(T_1, \dots, T_n)$.

3 Statements

$$\frac{\Gamma \vdash e : T \quad x : T \in \Gamma}{\Gamma \vdash x := e \text{ ok}} \text{T-ASSIGN}$$

$$\frac{\Gamma \vdash e : \text{bool} \quad \Gamma \vdash s^* \text{ ok}}{\Gamma \vdash \text{if } e \{s^*\} \text{ ok}} \text{T-IF}$$

$$\frac{\Gamma \vdash e : \text{bool} \quad \Gamma \vdash s_1^* \text{ ok} \quad \Gamma \vdash s_2^* \text{ ok}}{\Gamma \vdash \text{if } e \{s_1^*\} \text{ else } \{s_2^*\} \text{ ok}} \text{T-IF-ELSE}$$

$$\frac{\Gamma \vdash e : T}{\Gamma \vdash \text{print } e \text{ ok}} \text{T-PRINT}$$

4 Operational Semantics

The operational semantics are given as a big-step evaluation relation:

$$\langle e, \sigma \rangle \rightarrow v$$

4.1 Base Values

$$\frac{}{\langle n, \sigma \rangle \rightarrow n} \text{E-INT} \qquad \frac{}{\langle \mathbf{b}, \sigma \rangle \rightarrow \mathbf{b}} \text{E-BOOL}$$

4.2 Variables

$$\frac{\sigma(x) = v}{\langle x, \sigma \rangle \rightarrow v} \text{E-VAR}$$

4.3 Operations

$$\frac{\langle e_1, \sigma \rangle \rightarrow n_1 \quad \langle e_2, \sigma \rangle \rightarrow n_2}{\langle e_1 + e_2, \sigma \rangle \rightarrow n_1 + n_2} \text{E-ARITHMETIC}$$

$$\frac{\langle e_1, \sigma \rangle \rightarrow b_1 \quad \langle e_2, \sigma \rangle \rightarrow b_2}{\langle e_1 \wedge e_2, \sigma \rangle \rightarrow b_1 \wedge b_2} \text{E-LOGICAL}$$

$$\frac{\langle e, \sigma \rangle \rightarrow b}{\langle !e, \sigma \rangle \rightarrow \neg b} \text{E-NOT}$$

$$\frac{\langle e_1, \sigma \rangle \rightarrow n_1 \quad \langle e_2, \sigma \rangle \rightarrow n_2}{\langle e_1 < e_2, \sigma \rangle \rightarrow (n_1 < n_2)} \text{E-COMPARISON}$$

$$\frac{\langle e_1, \sigma \rangle \rightarrow v_1 \quad \langle e_2, \sigma \rangle \rightarrow v_2}{\langle e_1 == e_2, \sigma \rangle \rightarrow (v_1 == v_2)} \text{E-EQUALITY}$$

4.4 Variants

$$\frac{\langle e_1, \sigma \rangle \rightarrow v_1 \quad \cdots \quad \langle e_n, \sigma \rangle \rightarrow v_n}{\langle C(e_1, \dots, e_n), \sigma \rangle \rightarrow C(v_1, \dots, v_n)} \text{E-VARIANT-CONSTRUCTOR}$$

4.5 Statements

Statement execution is defined as:

$$\frac{\langle s, \sigma \rangle \rightarrow \sigma' \quad \langle e, \sigma \rangle \rightarrow v}{\langle x := e, \sigma \rangle \rightarrow \sigma[x \mapsto v]} \text{E-ASSIGN}$$

$$\frac{\langle e, \sigma \rangle \rightarrow \mathbf{true} \quad \langle s^*, \sigma \rangle \rightarrow \sigma'}{\langle \mathbf{if } e\{s^*\}, \sigma \rangle \rightarrow \sigma'} \text{E-IF-TRUE}$$

$$\frac{\langle e, \sigma \rangle \rightarrow \mathbf{false}}{\langle \mathbf{if } e\{s^*\}, \sigma \rangle \rightarrow \sigma} \text{E-IF-FALSE}$$

$$\frac{\langle e, \sigma \rangle \rightarrow \mathbf{true} \quad \langle s_1^*, \sigma \rangle \rightarrow \sigma'}{\langle \mathbf{if } e\{s_1^*\} \text{ else } \{s_2^*\}, \sigma \rangle \rightarrow \sigma'} \text{E-IF-ELSE-TRUE}$$

$$\frac{\langle e, \sigma \rangle \rightarrow \mathbf{false} \quad \langle s_2^*, \sigma \rangle \rightarrow \sigma'}{\langle \mathbf{if } e\{s_1^*\} \text{ else } \{s_2^*\}, \sigma \rangle \rightarrow \sigma'} \text{E-IF-ELSE-FALSE}$$